



MoodCloset – Your Smart Digital Closet

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1. INTRODUCTION

1.1 Purpose

The purpose of the MoodCloset application is to provide users with an intelligent digital wardrobe, allowing them to organize their clothing, receive AI mood-based outfit recommendations, detect their skin undertone for color matching, and share their looks on a social feed.

This SRS guides developers, designers, testers, and stakeholders by defining system behavior, constraints, and technical requirements.

1.2 Scope

MoodCloset is a mobile application that:

- Digitizes a user's wardrobe using uploaded images
- Generates outfits based on mood emojis
- Offers skin-undertone color recommendations
- Allows sharing final looks with the community
- Supports AI-assisted styling and personalization

System Goals

- Provide AI-powered mood-based outfit recommendations
- Offer digital wardrobe management
- Recommend color palettes using undertone selection
- Provide a social platform for sharing outfits
- Ensure fast, stable, and user-friendly interactions

System Benefits

- Reduces decision fatigue
- Helps users discover their personal style
- Organizes all clothing items digitally
- Promotes community engagement and inspiration
- Increases user confidence through color-matching features

System Boundaries

Included

- Wardrobe digitization
- Mood-based AI outfit generation
- Undertone-based color palette suggestions
- Outfit-sharing feed
- Basic profile & settings management

Excluded

- Real-time shopping integration
- AR try-on
- Advanced skin-tone detection using ML (manual selection only)

1.3 Definitions and Acronyms

Term	Definition
AI	Artificial Intelligence used for outfit generation
Mood Engine	Rule-based engine mapping mood → outfit style
Undertone	Warm, Cool, or Neutral skin undertone
Wardrobe Item	Clothing uploaded into app
Feed	Social page showing user outfits

1.4 Competitor Analysis

Competitor	Strengths	Weaknesses
Stylebook	Excellent wardrobe mgmt	No AI, no mood styling
Cladwell	Good outfit suggestions	No social feed, limited personalization
Fashable	Strong AI features	Weak wardrobe organization
Pinterest	Strong inspiration library	No wardrobe or mood engine

- Mood-based outfit recommendations
- Integrated wardrobe + AI + undertone engine
- Social feed for outfit sharing
- Personalized mood and color analytics
- Faster, simpler, aesthetic interface for Gen-Z

2. OVERALL DESCRIPTION

2.1 System Context

MoodCloset interacts with:

- User (uploads images, chooses mood, selects undertone)
- AI Mood Engine (generates outfit rules)
- Image processing module (categorizes clothing)
- Database (stores wardrobe, posts, profiles)
- Notification service

2.2 User Characteristics (Personas)

Persona 1: The Fashion Lover





Persona 3: The Social Sharer



2.3 Technical Constraints

- Requires stable internet for uploading images
- Image recognition limited to rule-based category detection
- AI depends on pre-defined mood→style mapping

Security Constraints

- All images encrypted at rest
- Authentication required
- No anonymous posting

Regulatory Constraints

- Must comply with general privacy standards
- All wardrobe photos considered user personal data

2.4 Assumptions & Dependencies

Assumptions

- Users upload real, clear images
- Users have smartphones with camera access
- App stores user undertone manually

Dependencies

- Cloud database storage
 - AI rule engine
 - Image processing library
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3. FUNCTIONAL REQUIREMENTS (FRs)

Req ID	Requirement Name	Description
RQ1	Upload Clothing Item	System shall allow users to upload clothing images from device gallery.
RQ2	Auto-Categorize Item	System shall auto-detect clothing type (top, bottom, etc.).
RQ3	Add Preloaded Item	System shall allow adding clothing from built-in library.
RQ4	View Wardrobe	System shall display wardrobe items in categories.
RQ5	Edit Clothing Item	Users shall edit clothing details (color, category).
RQ6	Delete Clothing Item	Users may delete items permanently.
RQ7	Detect Undertone	System shall allow users to choose undertone.
RQ8	Recommend Palette	System shall recommend colors based on undertone.
RQ9	Select Mood	User selects mood emoji (happy, calm, edgy).
RQ10	Generate Outfit	System shall produce outfit suggestions.
RQ11	Display Final Look	Generated outfits shall be displayed in <2 seconds.
RQ12	Share Outfit	Users shall post final looks on feed.
RQ13	Like/Comment	Users can like and comment on posts.
RQ14	Save Post	Users may save outfits for later.
RQ15	Create Account	System shall allow registration & login.
RQ16	Session Security	Sessions expire after inactivity.

4. NON-FUNCTIONAL REQUIREMENTS (NFRs)

NFR ID	Category	Requirement
NFR-1	Performance	Wardrobe loads in <3s
NFR-2	Performance	Outfit recommendation <2s
NFR-3	Security	Passwords hashed & encrypted
NFR-4	Usability	2-click access to main features
NFR-5	Accessibility	Supports dark mode & readable fonts
NFR-6	Reliability	99% uptime
NFR-7	Scalability	Supports thousands of wardrobe images
NFR-8	Maintainability	Modular code structure

5. SYSTEM FEATURES / MODULES

Feature 1. Wardrobe Management Module

Inputs: Clothing images

Processing: Categorization, save item

Outputs: Organized wardrobe grid

Feature 2. Mood-Based Outfit Engine

Inputs: Selected mood

Processing: Rule-based outfit creation

Outputs: Outfit set

Feature 3. Undertone Color Module

Inputs: Undertone selection

Outputs: Recommended palette

Feature 4. Social Feed Module

Inputs: Outfit post

Outputs: Post displayed, likes/comments

Feature 5. Authentication Module

Handles: Login, sessions, profile settings

6. EXTERNAL INTERFACES

The MoodCloset system interacts with several external interfaces to support wardrobe digitization, AI-based outfit generation, and social sharing. This section describes all user-facing, hardware, software, and communication interfaces required for the system to operate.

6.1 User Interfaces

MoodCloset includes multiple user interfaces designed for simplicity, quick interaction, and visually appealing styling.

Wardrobe Interface

- Allows users to upload clothing images.
- Displays categorized clothing items (tops, bottoms, accessories).
- Provides sorting features (color, category, season).

Mood Selection Interface

- Shows mood emojis selectable by the user.
- Sends selected mood to the AI engine to generate outfit recommendations.

Outfit Recommendation Screen

- Displays single or multiple outfits suggested by the AI.
- Shows matching colors and mood-inspired combinations.

Undertone Detection Screen

- Allows users to manually select undertone (Warm, Cool, Neutral).
- Displays recommended color palette.

Social Feed Interface

- Allows users to share outfit posts.
- Displays likes, comments, and saves.
- Similar layout to Instagram Explore but restricted to outfits only.

Profile & Settings UI

- Displays user wardrobe statistics, preferences, and saved looks.
- Provides toggles for privacy settings, notifications, and linked accounts.

6.2 Hardware Interfaces

Although MoodCloset is designed primarily for mobile devices, it interacts with the following hardware components:

Mobile Device Camera

- Used for capturing clothing images.
- Ensures clear, well-lit photos for AI recognition.

- Temporarily stores uploaded images before sending to cloud storage.

Smartphone Sensors (Optional)

- Ambient light sensor for better auto-enhancement of clothing images.

Cloud Storage Servers

- Used for long-term storage of wardrobe images.
- Must support high-resolution images with optimized compression.

6.3 Software Interfaces

MoodCloset integrates with multiple APIs and backend services to ensure accurate recommendations and real-time outfit generation.

AI Outfit Engine API

- Receives mood, wardrobe data, and color palettes.
- Returns outfit combinations in structured JSON format.

Image Recognition API

- Used to detect clothing type, color, and features.
- Processes uploaded images and sends metadata to the wardrobe database.

Color Palette Analysis Module

- Matches clothing colors to undertone rules.
- Generates “warm,” “cool,” or “neutral” recommended palettes.

Database Interface

- Stores wardrobe items, posts, user preferences, and uploaded images.
- Must support CRUD operations and real-time updates.

Authentication API

- Allows login, sign-up, and secure session management.
- Supports OAuth2 and encrypted token handling.

- Manages posts, likes, comments, and user interactions.
- Connects with database and notification services.

6.4 Communication Interfaces

MoodCloset uses standard communication protocols required for stable, secure, and fast interaction between users, servers, and third-party APIs.

HTTPS (TLS 1.2+)

- Used for all communication between app and backend.
- Ensures encrypted transfer of images, login data, and personal info.

RESTful API Endpoints

- Handle CRUD operations for wardrobe items, posts, and recommendations.
- Return JSON responses for smooth mobile integration.

WebSockets (for real-time updates)

- Used for:
 - Live comment updates
 - Real-time outfit feed refresh
 - Notification delivery

TCP/IP

- Ensures reliable, stable network communication at the device level.

Cloud Messaging Services

- Used for push notifications (likes, comments, new AI recommendations).
- Supports Apple Push Notification Service (APNs) and Firebase (FCM).

UC1 – Manage Wardrobe

Primary Actor: User

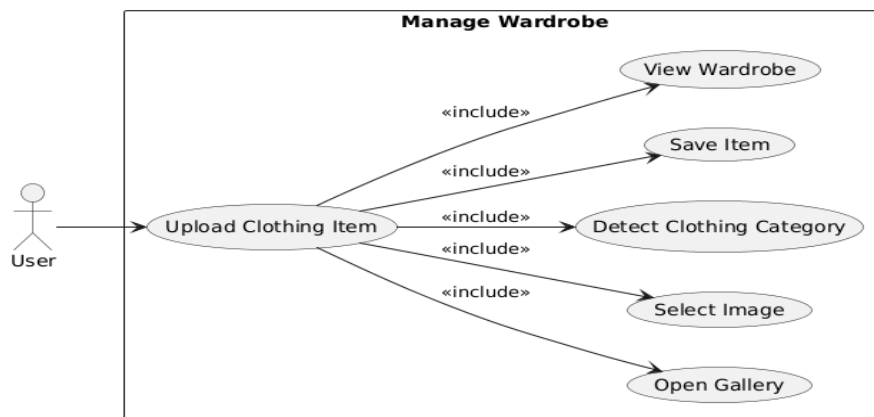
Preconditions: User is logged in; wardrobe DB exists

Postconditions: Wardrobe updated

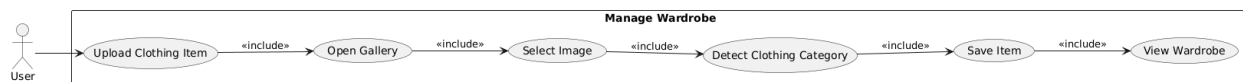
Main Flow

1. User selects “Upload Clothing Item”
2. System opens gallery
3. User selects image
4. System detects clothing category
5. System saves item
6. User views item in wardrobe

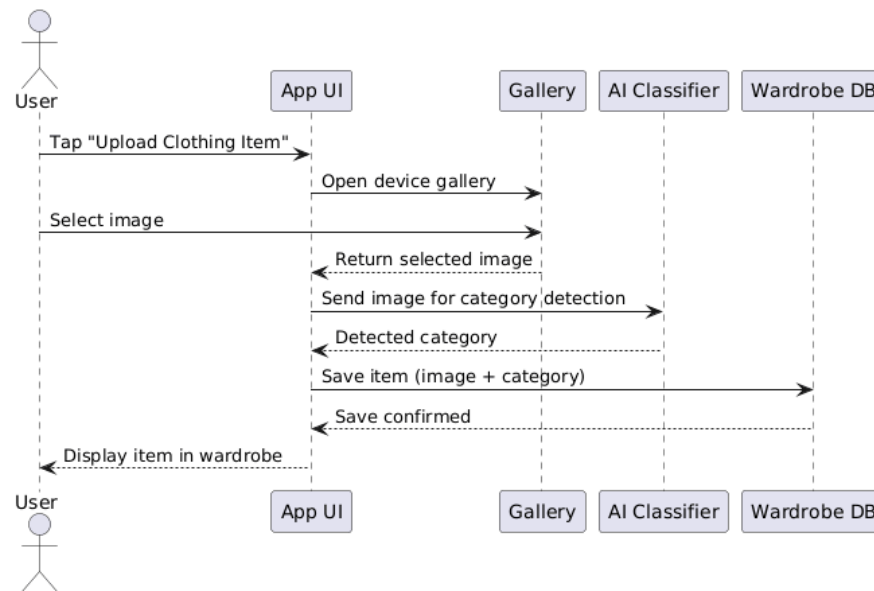
Use Case Diagram - Upload Clothing Item (Detailed)



UC1 - Upload Clothing Item



Sequence Diagram - UC1 Upload Clothing Item



UC2 – Detect Undertone & Recommend Palette

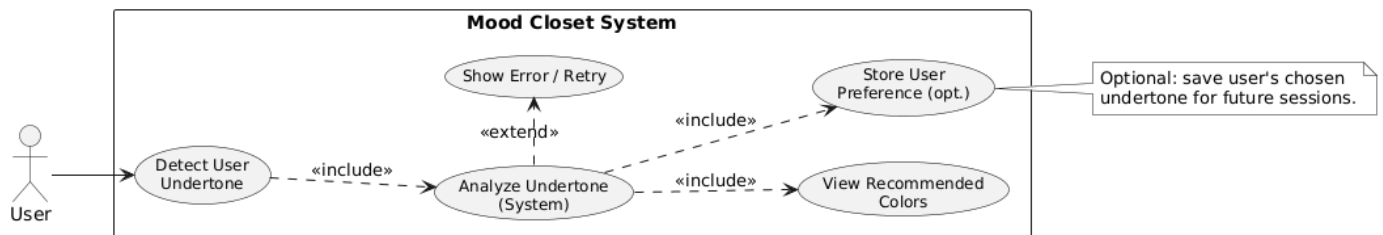
Primary Actor: User

Preconditions: Undertone selection available

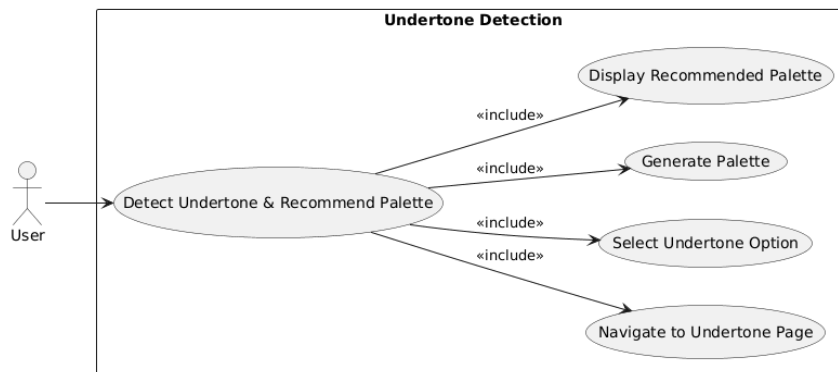
Postconditions: Palette displayed

Main Flow

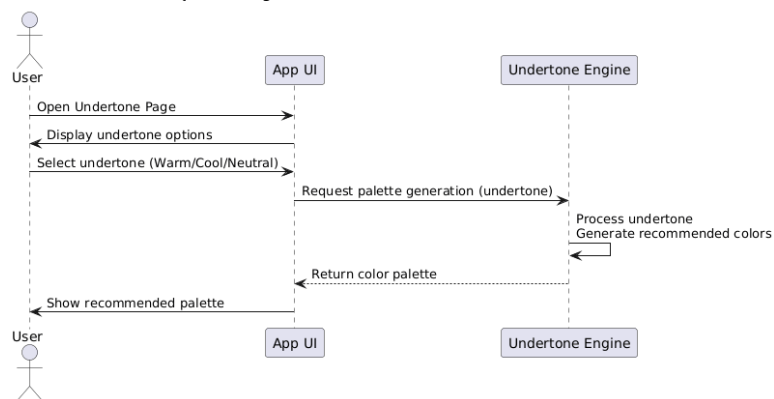
1. User navigates to Undertone page
2. Selects undertone
3. System generates palette
4. Palette displayed



Use Case Diagram - Detect Undertone & Recommend Palette



Sequence Diagram - Detect Undertone & Recommend Palette



UC3 – Get Outfit Recommendation

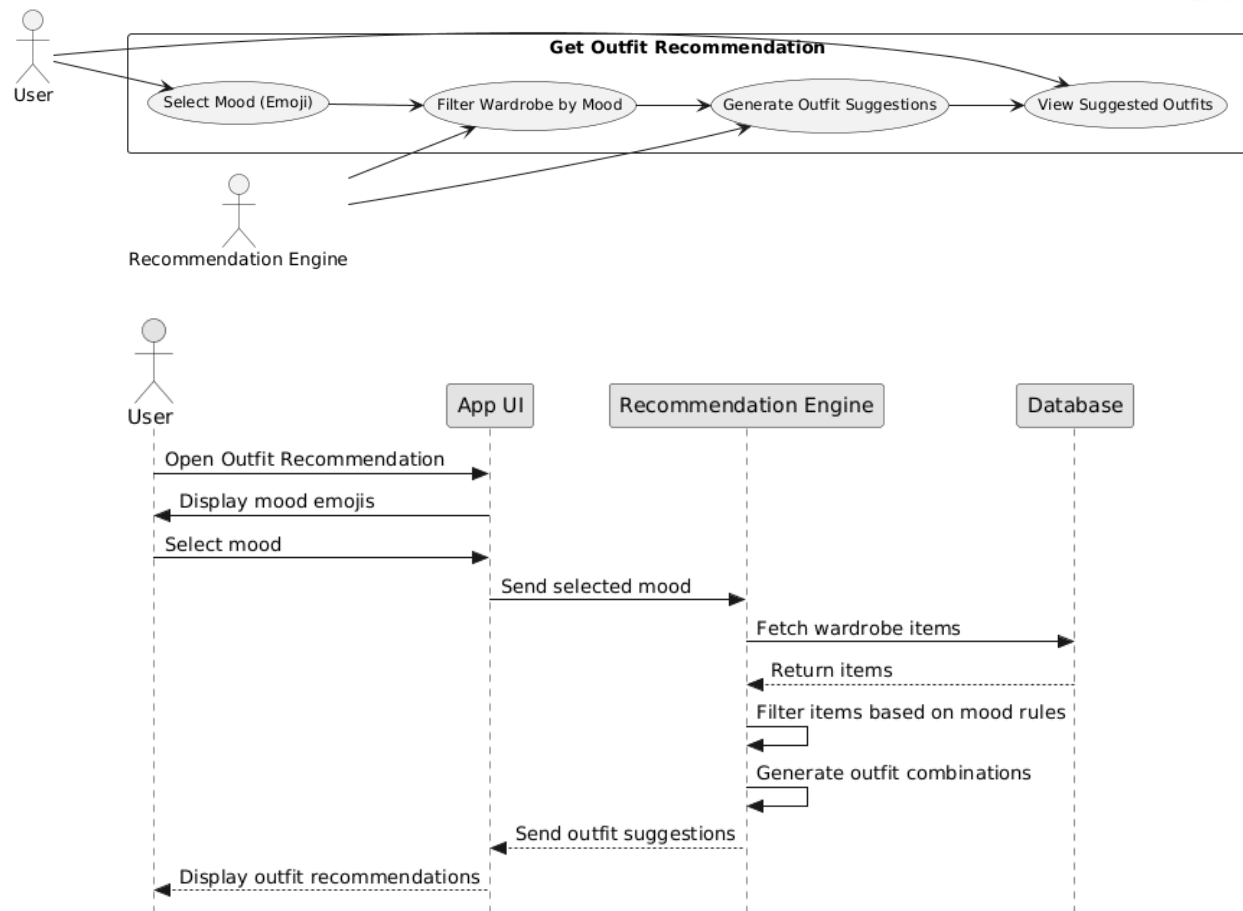
Primary Actor: User

Preconditions: Wardrobe has items

Postconditions: Outfit displayed

Main Flow

1. User opens “Get Outfit Recommendation”
2. Selects mood
3. System filters wardrobe
4. Generates outfit
5. Displays result



UC4 – Share Outfit

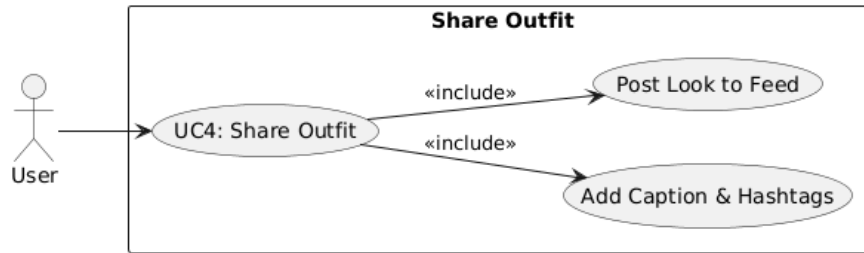
Primary Actor: User

Preconditions: User has a generated outfit

Postconditions: Look posted to feed

Main Flow

1. User selects “Share”
2. Adds caption/hashtags
3. System posts to feed



TRACEABILITY MATRIX

Req ID	UC1	UC2	UC3	UC4
RQ1	✓			
RQ2	✓			
RQ3	✓			
RQ4	✓			
RQ5	✓			
RQ6	✓			
RQ7		✓		
RQ8		✓		
RQ9			✓	
RQ10			✓	
RQ11			✓	
RQ12				✓
RQ13				✓
RQ14				✓

RQ15	✓			
RQ16	✓			

8. VALIDATION & ACCEPTANCE CRITERIA

Req ID	Validation Method	Acceptance Criteria
RQ1	UI Test	Image uploads successfully
RQ2	AI test	Category detected correctly
RQ7	Functional test	Undertone saved
RQ10	Performance test	Outfit <2s
RQ11	UI test	Outfit displayed cleanly
RQ12	Functional test	Post visible in feed
NFR-1	Load test	Wardrobe loads <3s

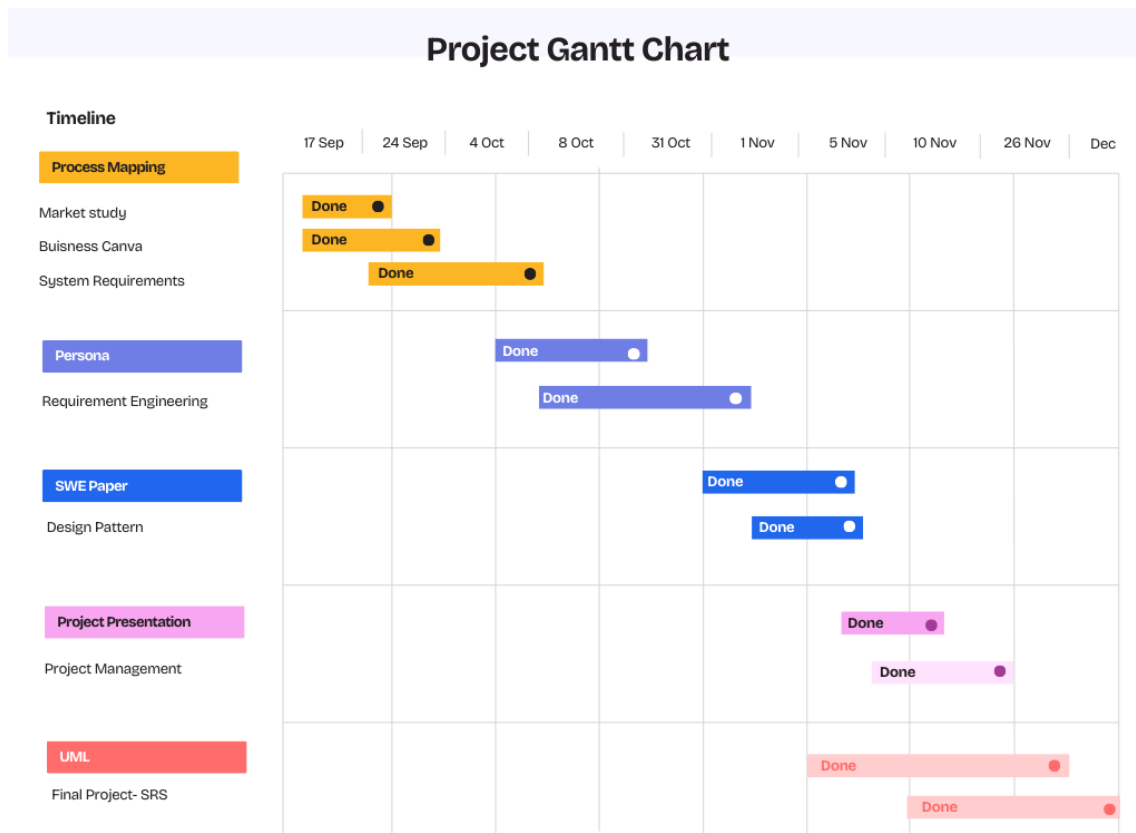
9. APPENDICES

9.1 Business Model Canvas

MoodCloset – Mood-Based Fashion & Outfit Sharing App



9.2 Gant Chart



9.3 References

